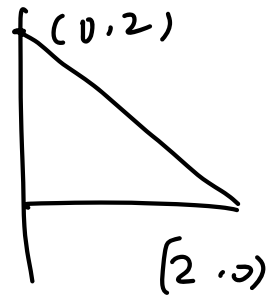


$$1. f(x,y) = 4y + x^2 - 2x + 1$$

$$f_x = 2x - 2 \quad f_y = 4$$



$$O_n \rightarrow 0 \leq x \leq 2 \quad y = 0$$

$$f(x, 0) = 4 \cdot 0 + x^2 - 2x + 1 = (x-1)^2$$

$$(0,0) \quad (1,0) \quad (2,0)$$

$$O_n \quad y = -x + 2, \quad 0 \leq x \leq 2$$

$$f(x, -x+2) = 4(-x+2) + x^2 - 2x + 1 = x^2 - 6x + 9 = (x-3)^2$$

$$(0,0) \quad (2,0)$$

$$O_n \quad x = 0 \quad 0 \leq y \leq 2$$

$$f(0, y) = 4y + 0^2 - 2 \cdot 0 + 1 = 4y + 1$$

$$(0,0) \quad (0,2)$$

$$\max = f(0,2) = 9$$

$$\min = f(1,0) = 0$$

$$2. \quad \begin{cases} \nabla f = \lambda \cdot \nabla g \\ g(x, y) = 0. \end{cases} \quad \lambda, x, y \quad g(x, y) \stackrel{\text{def}}{=} x^2 + y^2 - 5$$

$$\nabla f = \langle 6x^2, -3y^2 \rangle \quad \nabla g = \langle 2x, 2y \rangle$$

$$\Rightarrow \begin{cases} 6x^2 = \lambda \cdot 2x \Rightarrow \lambda = 3x \\ -3y^2 = \lambda \cdot 2y \Rightarrow \lambda = -\frac{3y}{2} \\ x^2 + y^2 - 5 = 0 \end{cases}$$

Case 1: $x \neq 0$ and $y \neq 0 \Rightarrow$ "usual".

Case 2: $x = 0$

Case 3: $y = 0$

Case 1: $3x = -\frac{3y}{2} \quad x = -\frac{y}{2}$
 $\left(-\frac{y}{2}\right)^2 + y^2 = 5 \Rightarrow \frac{5}{4}y^2 = 5 \Rightarrow y = \pm 2$

$$(x, y) = (-1, 2) \text{ or } (1, -2)$$

Case 2: $\begin{cases} 6x^2 = \lambda \cdot 2x \\ -3y^2 = \lambda \cdot 2y \\ x^2 + y^2 - 5 = 0 \end{cases} \Rightarrow \begin{cases} -3y^2 = \lambda \cdot 2y \\ y^2 - 5 = 0 \end{cases}$

$$y = \pm \sqrt{5} \Rightarrow (x, y) = (0, \sqrt{5}) \text{ or } (0, -\sqrt{5})$$

Case 3: $(x, y) = (\sqrt{5}, 0) \text{ or } (-\sqrt{5}, 0)$

$$\text{max} = f(\sqrt{5}, 0) = 10\sqrt{5}$$

$$\max = f(\sqrt{5}, 0) = 10\sqrt{5}$$

$$\min = f(-\sqrt{5}, 0) = -10\sqrt{5}$$

$$3. \quad f(x, y) = x - 3y$$

$$g(x, y) = x^2 + 2y^2 - 3$$

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 0 \end{cases}$$

$$\begin{cases} \langle 1, -3 \rangle = \lambda \langle 2x, 4y \rangle \\ x^2 + 2y^2 - 3 = 0 \end{cases}$$

$$2\lambda x = 1 \quad 4\lambda y = -3$$

$$\left(\frac{1}{2\lambda}\right)^2 + 2 \cdot \left(\frac{-3}{4\lambda}\right)^2 - 3 = 0$$

$$\frac{1}{4\lambda^2} + 2 \cdot \frac{9}{16\lambda^2} - 3 = 0$$

$$4 + 18 - 48\lambda^2 = 0$$

$$\lambda^2 = \frac{22}{48} = \frac{11}{24}$$

$$\left(\frac{1}{4} + \frac{9}{8}\right)\frac{1}{\lambda^2} = 3 \quad \lambda = \pm \sqrt{\frac{11}{24}}$$

$$\lambda = \sqrt{\frac{11}{24}}, \quad x = \frac{1}{2\lambda} = \frac{1}{2} \sqrt{\frac{24}{11}}$$

$$x - 3y = \frac{11}{4} \sqrt{\frac{24}{11}} \quad y = -\frac{3}{4\lambda} = -\frac{3}{4} \sqrt{\frac{24}{11}}$$

$$\lambda = -\sqrt{\frac{11}{24}}, \quad x = -\frac{1}{2} \sqrt{\frac{24}{11}} \\ y = \frac{3}{4} \sqrt{\frac{24}{11}}$$

$$x - 3y = \left(-\frac{1}{2} - \frac{9}{4}\right)\left(-\frac{1}{\lambda}\right) \\ = -\frac{11}{4} \sqrt{\frac{24}{11}}$$

$$\max = f\left(\frac{1}{2} \sqrt{\frac{24}{11}}, -\frac{3}{4} \sqrt{\frac{24}{11}}\right) = \frac{11}{4} \sqrt{\frac{24}{11}}$$

$$\min = f\left(-\frac{1}{2} \sqrt{\frac{24}{11}}, \frac{3}{4} \sqrt{\frac{24}{11}}\right) = -\frac{11}{4} \sqrt{\frac{24}{11}}$$

$$4. f(x, y) = x^2 y - xy + x$$

$$0 \leq x \leq 2 \quad -2 \leq y \leq 0$$

$$f_x = 2xy - y + 1 = 0$$

$$f_y = x^2 - x = 0$$

$$\begin{cases} x = 1 \\ y = -1 \end{cases}$$

$$\begin{cases} x = 0 \\ y = 1 \end{cases}$$

$$f_{xx} = 2y \quad f_{xy} = 2x - 1$$

$$f_{yy} = 0$$

$$\begin{pmatrix} 2y & 2x-1 \\ 2x-1 & 2 \end{pmatrix} \quad (1, -1)$$

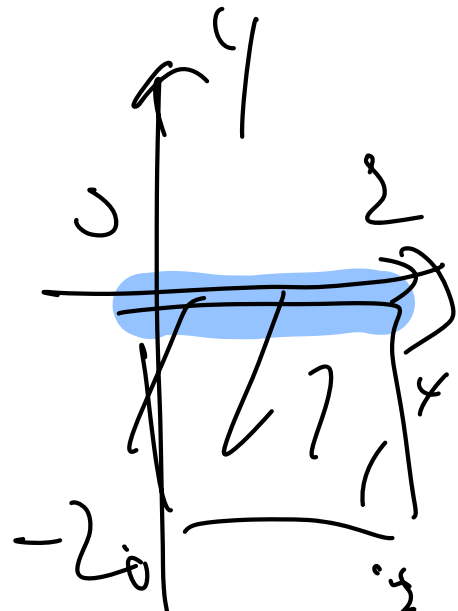
$$= \begin{pmatrix} -2 & 1 \\ 1 & 0 \end{pmatrix} \quad D = -1 < 0 \\ \text{saddle}$$

$$\begin{pmatrix} 2y & 2x-1 \\ 2x-1 & 2 \end{pmatrix} \quad (0, 1)$$

$$= \begin{pmatrix} 2 & -1 \\ -1 & 0 \end{pmatrix} \quad D = -1 < 0 \\ \text{saddle}$$

$$f(x, y) = x^2 y - xy + x$$

$$0 \leq x \leq 2 \quad -2 \leq y \leq 0$$



$$0 \leq x \leq 2 \quad (x, y) =$$

$$f(x, 0) = x \quad (0, 0) \quad (2, 0)$$

0 2

$$0 \leq x \leq 2$$

$$f(x, -2) = -2x^2 + 2x + y$$

$$= -2x^2 + 3x$$

$$= -2\left(x^2 - \frac{3}{2}x + \left(\frac{3}{4}\right)^2\right) + 2\left(\frac{3}{4}\right)^2$$

$$(x, y) = (0, -2) \quad \left(\frac{3}{4}, -2\right) \quad (2, -2)$$

0 1 8

$$-2 \leq y \leq 0$$

$$f(0, y) = 0$$

0

$$-2 \leq y \leq 0$$

$$f(2, y) = 4y - 2y + 2 = 2y + 2$$

$$(x, y) = (2, -2), (2, 0)$$

-2 0

$$\max f\left(\frac{3}{4}, -2\right) = \frac{8}{5}$$

$$\min f(2, -2) = -2$$

5.

$$(1, 3) : \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} D = -3 < 0$$

saddle

$$f(1, 0) : \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} D = 5 > 0$$

$f_{xx} > 0$

min